

Use Of New Materials In SoC Manufacture

Apart from updating the architecture of chips, developers have been trying to find new alloys and elements which would further improve performance, while at the same time reducing cost of production. One such alloy, which is being used currently, is a mixture of silicon-germanium (SiGe).

Since 2000 or so, SiGe has been widely used to enhance the performance of ICs of various types.

SiGe can be processed on equipment in nearly the same way as ordinary silicon. SiGe doesn't have some of the drawbacks of III-V compound semiconductors like gallium arsenide (GaAs). For example, it doesn't lack a native oxide (important for forming MOS structures) and doesn't suffer from mechanical fragility, which limits the wafer size of GaAs.

This results in costs that are only a small multiple of ordinary silicon, and so much lower than competing technologies like GaAs.

SiGe allows for two main improvements compared to ordinary silicon:

First, adding Germanium increases the lattice constant of the alloy. If a layer of Si is grown on top of SiGe, there will be mechanical strain induced by the lattice's constant mismatch. The strained layer will have a higher carrier mobility than an unstrained Si. This can be used, for example, to balance the performance of PMOS and NMOS transistors, reducing the area needed for a given CMOS circuit.

Secondly, the SiGe alloy can be used selectively in the base region of a BJT to form a heterojunction bipolar transistor (HBT). SiGe HBT's have been demonstrated with speeds (fT) up to 500 GHz, and are commercially available with fT up to 240 GHz. The SiGe HBT also makes less noise than a standard silicon BJT.

The future of chips could also lead developers to completely abandon silicon in favour of Germanium. Researchers claim that silicon has become difficult to work with in this age of miniaturization. In a bid to create even smaller processes, Germanium in itself may present the answer. The element has superb electrical properties and innovative techniques have made it possible for companies to work with Germanium.

Upcoming State Of The Art SoC

In 2017, the SoC market has been abuzz with reports of the Qualcomm Snapdragon 835 chipset and the Exynos 8895 SoC, both of which are housed in Samsung's Galaxy S8 device. Both are based on the 10nm FinFET architecture.